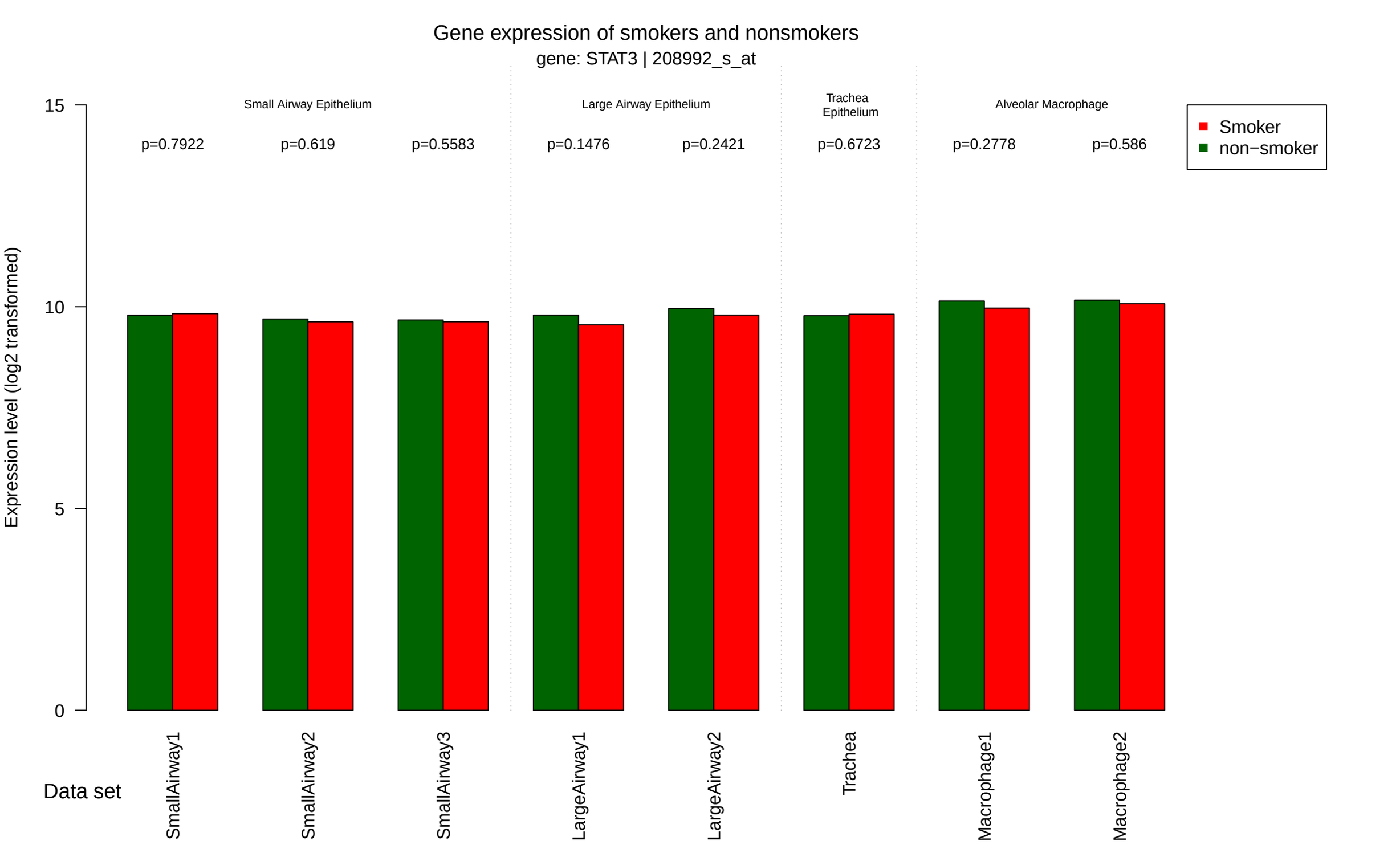
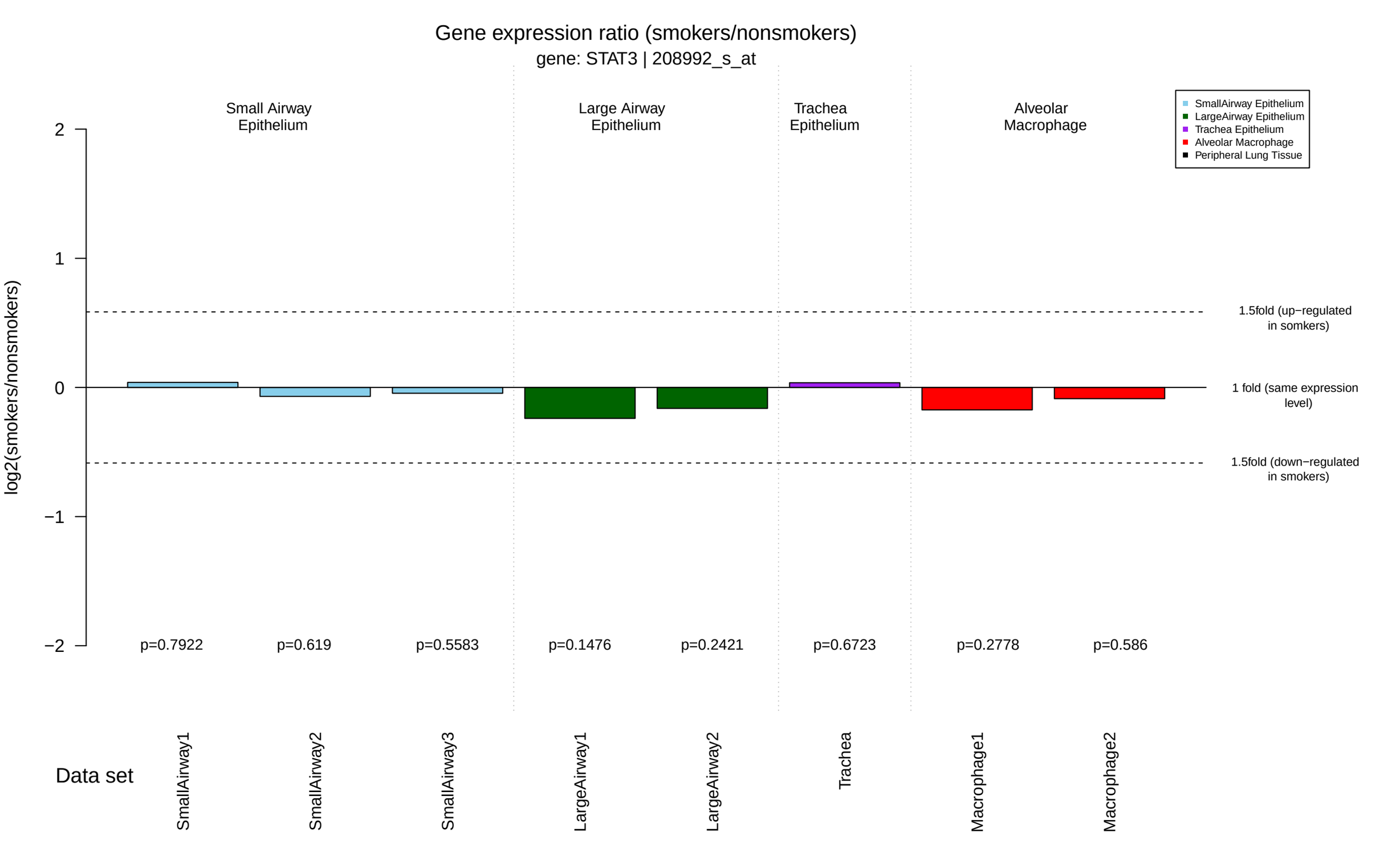
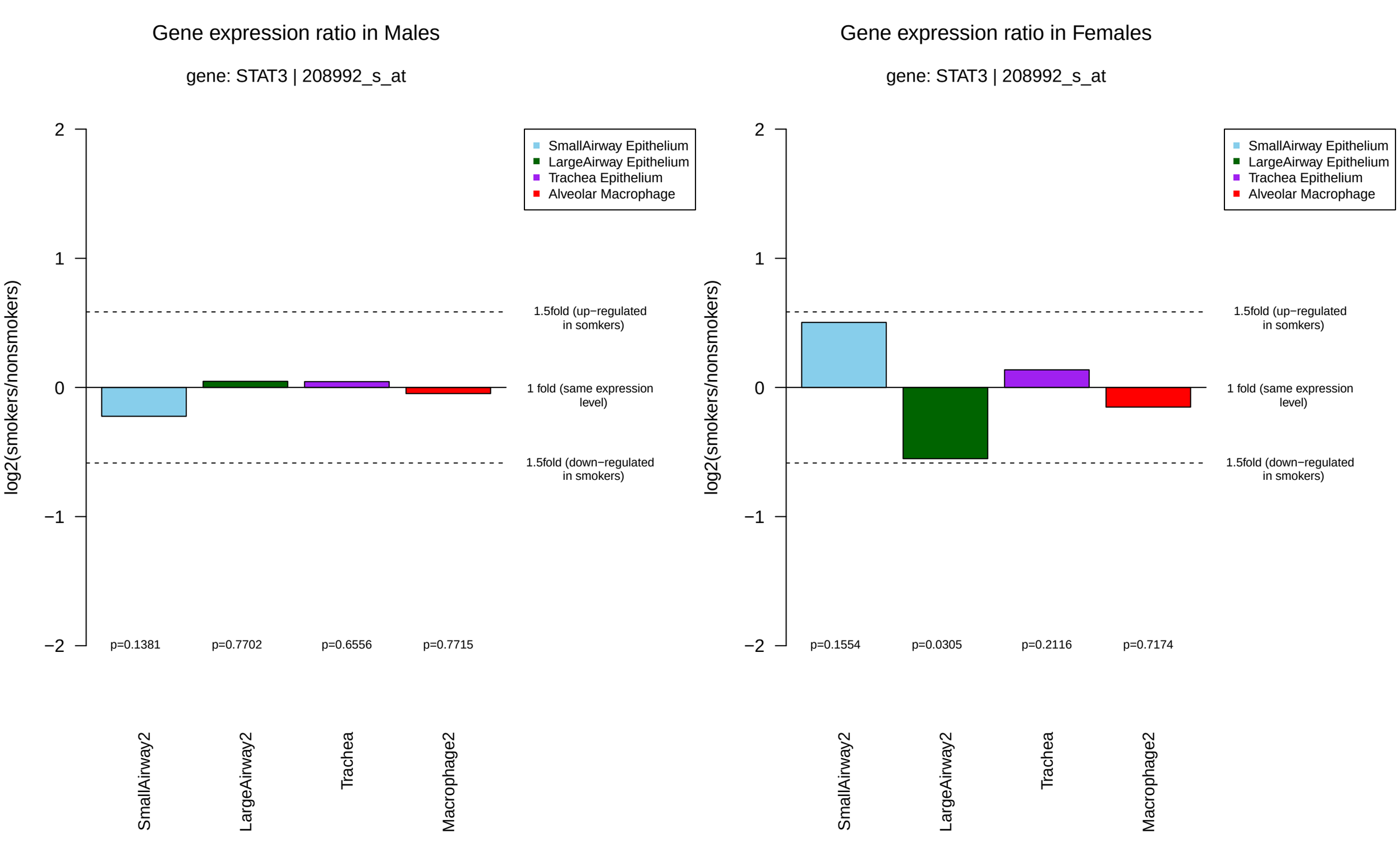
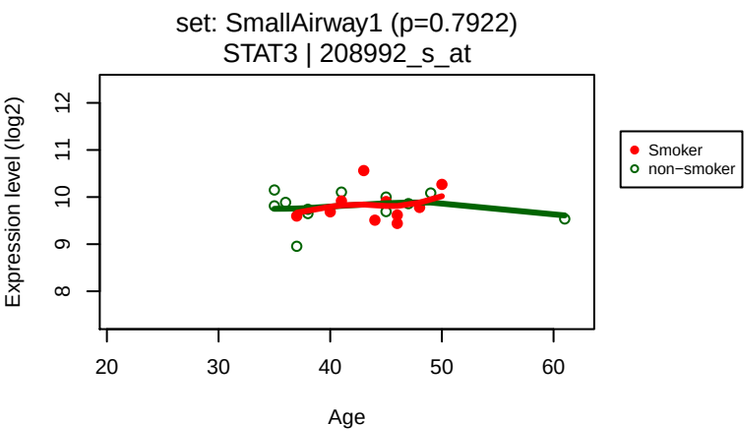
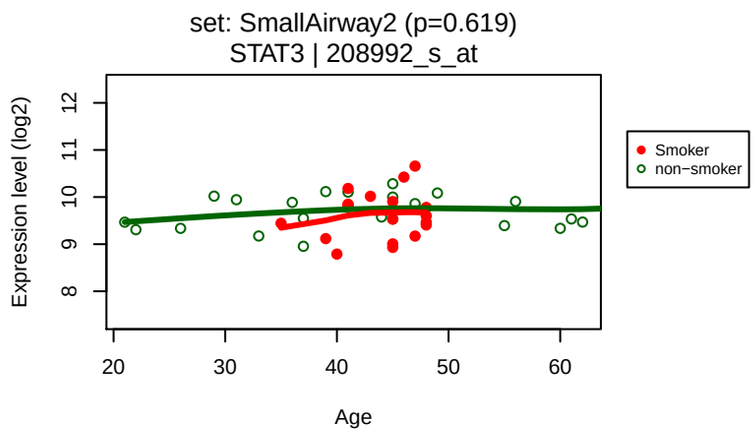
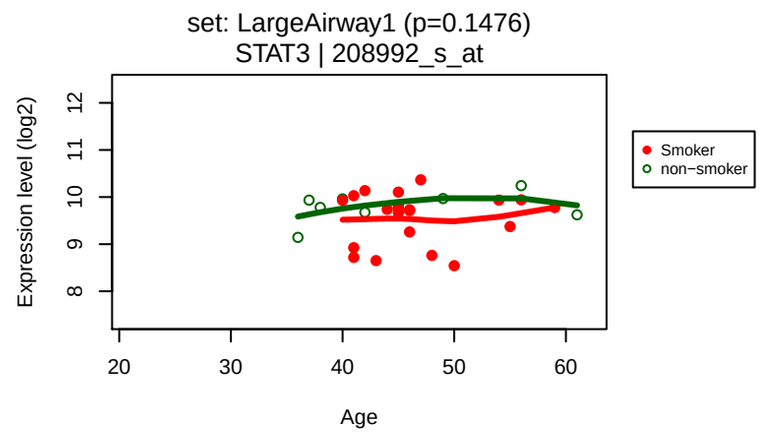

Gene expression of smokers and nonsmokers
gene: STAT3 | 208992_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7922
p=0.619
p=0.5583
p=0.1476
p=0.2421
p=0.6723
p=0.2778
p=0.586
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

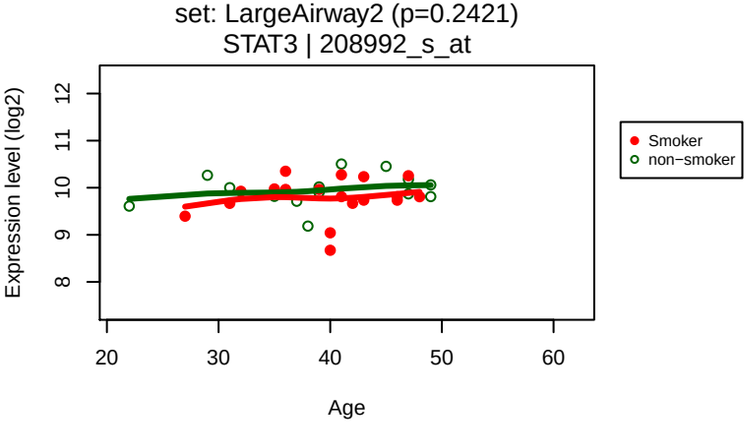

Gene expression ratio (smokers/nonsmokers)
gene: STAT3 | 208992_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7922
p=0.619
p=0.5583
p=0.1476
p=0.2421
p=0.6723
p=0.2778
p=0.586
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

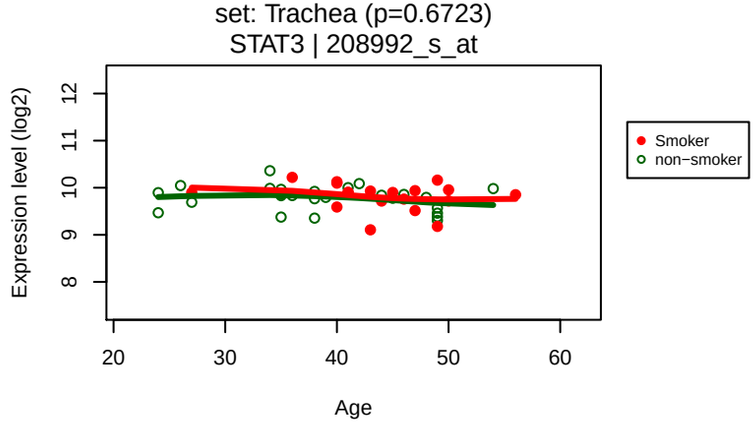

Gene expression ratio in Males
gene: STAT3 | 208992_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1381
p=0.7702
p=0.6556
p=0.7715
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: STAT3 | 208992_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1554
p=0.0305
p=0.2116
p=0.7174
SmallAirway2
LargeAirway2
Trachea
Macrophage2

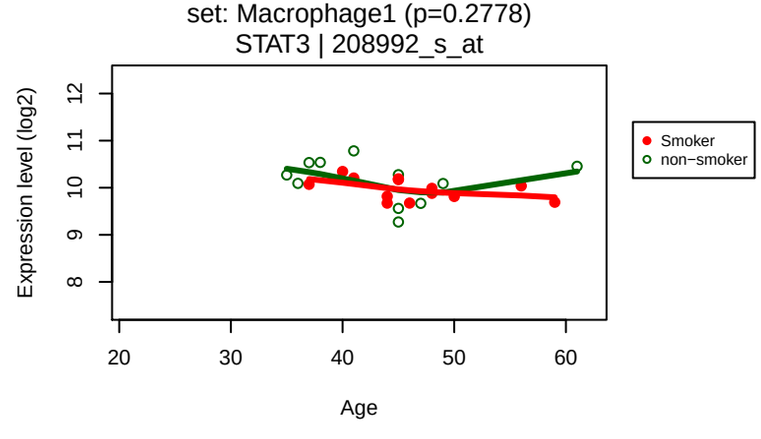

set: SmallAirway1 (p=0.7922)
STAT3 | 208992_s_at
Expression level (log2)
Age
Smoker
non−smoker

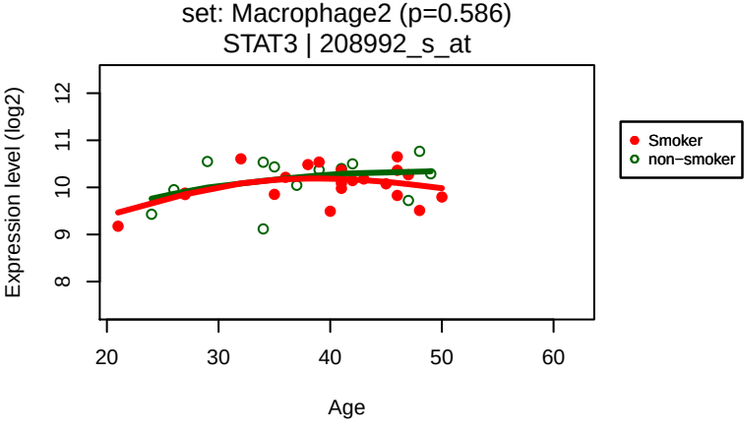

set: SmallAirway2 (p=0.619)
STAT3 | 208992_s_at
Expression level (log2)
Age
Smoker
non−smoker

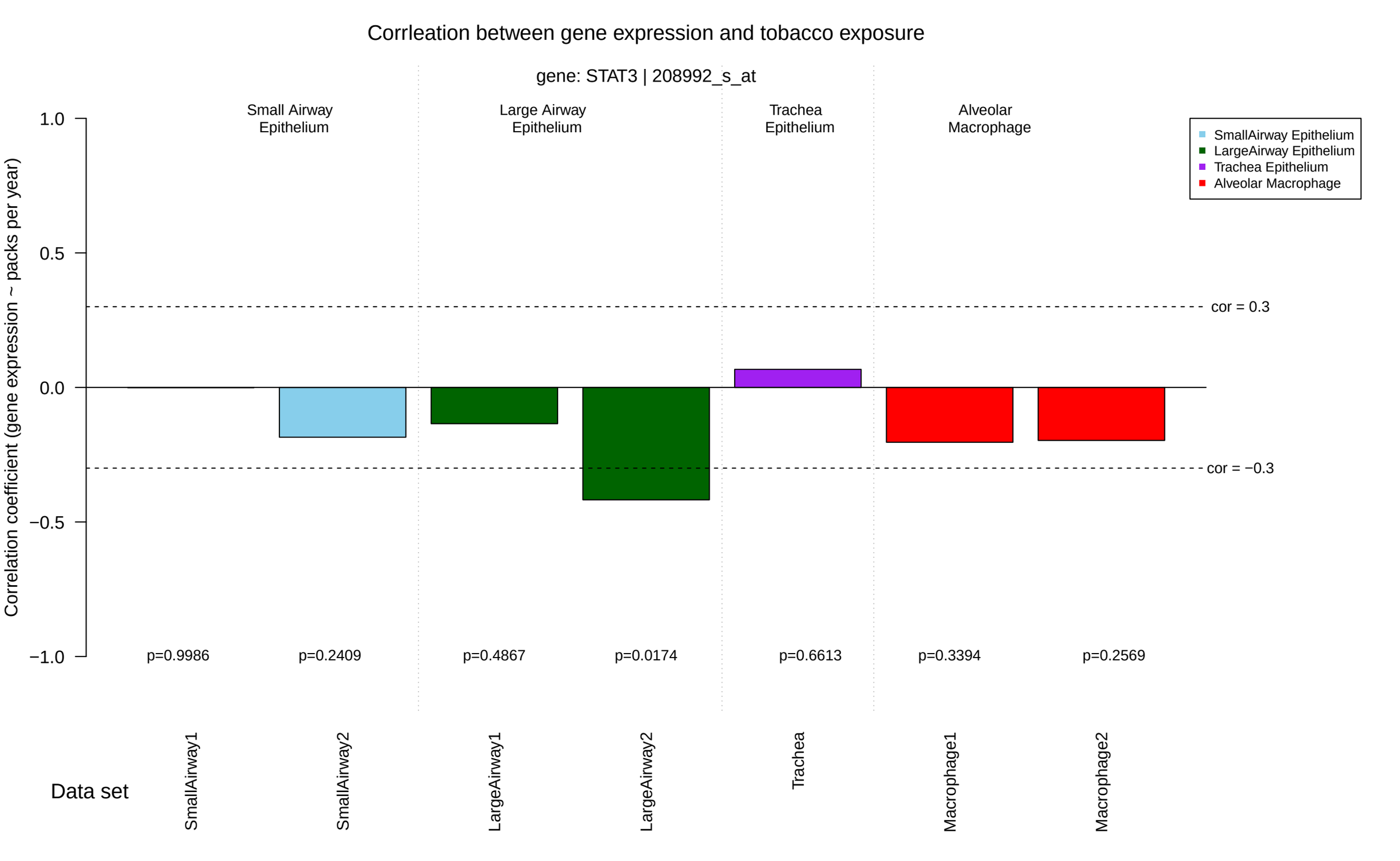

set: LargeAirway1 (p=0.1476)
STAT3 | 208992_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2421)
STAT3 | 208992_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6723)
STAT3 | 208992_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2778)
STAT3 | 208992_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.586)
STAT3 | 208992_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: STAT3 | 208992_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9986
p=0.2409
p=0.4867
p=0.0174
p=0.6613
p=0.3394
p=0.2569
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2